

Question		Answer	Mark	Mark scheme	Additional guidance
9	(a)	4.52×10^3	M1	for $2.04 \dots \times 10^7$ oe eg $2.04 \dots \times 10^{-5} \div 10^{-12}$ or $20.4 \dots \times 10^6$ or 204(08163.27) or for correct value of T , 4517.(53....), not written in standard form, eg 4520	May be given correct to 3 sig figs or more
			A1	for answer in the range 4.51×10^3 to 4.52×10^3 (SC B1 for $6.32 \dots \times 10^{-1}$)	
	(b)	Explanation	M1	for method to find the scale factor or decreased value in T , eg $\sqrt{\frac{1.1}{1.05^3}}$ (= 0.97.....) oe or $\sqrt{\frac{5.6 \times 10^{-5} \times 1.1}{(1.4 \times 10^{-4} \times 1.05)^3}}$ (= $4.40 \dots \times 10^3$) oe	Award mark for a correct method to calculate the scale factor or the percentage increases in w and d^3 or the decreased value of T
			C1	(dep M1) for explanation eg value of scale factor less than 1, so a decrease in T OR compares $4.40 \dots \times 10^3$ with their value of T from (a) provided answer to (a) is greater	This mark may only be awarded if supported by numerical evidence